

Film Festival Database

Relational Model and Relational Algebra

1 Relational Model

Festival(id, title, year, location)

Category(id, name)

Person(id, name, type)

Staff(id, name, type, specialization)

Film(id, title, duration, year, country, category__id, final__score)

Screening(id, date, time, room, capacity, film__id, coordinator__id)

FestivalCategory(festival__id, category__id)

FestivalFilm(festival__id, film__id)

FilmDirector(film__id, director__id)

FilmActor(film__id, actor__id)

Evaluation(judge__id, film__id, score)

Director = $\pi_{id, name}(\sigma_{type='director'}(Person))$

Actor = $\pi_{id, name}(\sigma_{type='actor'}(Person))$

Judge = $\pi_{id, name, specialization}(\sigma_{type='judge'}(Staff))$

Coordinator = $\pi_{id, name}(\sigma_{type='coordinator'}(Staff))$

2 Relational Algebra Queries

1. Films in a specific festival

$$\pi_{\text{Film.*}} (\text{Film} \bowtie_{\text{Film.id}=\text{FestivalFilm.film_id}} (\sigma_{\text{festival_id}=x} (\text{FestivalFilm})))$$

2. Number of films in each category

$$\text{id, name} \bowtie \text{COUNT Film.id AS film_count} (\text{Category} \bowtie_{\text{Category.id}=\text{Film.category_id}} \text{Film})$$

3. Directors of a film

$$\pi_{\text{name}} (\text{Director} \bowtie_{\text{Director.id}=\text{FilmDirector.director_id}} \sigma_{\text{film_id}=x} (\text{FilmDirector}))$$

4. Actors of a film

$$\pi_{\text{name}} (\text{Actor} \bowtie_{\text{Actor.id}=\text{FilmActor.actor_id}} \sigma_{\text{film_id}=x} (\text{FilmActor}))$$

5. Future screenings of a film

$$\begin{aligned} \mathcal{T}_{\text{date, time}} \Big(\pi_{\text{title, time, date, room, capacity, name}} \Big(\sigma_{\text{Film.id}=x \wedge \text{datetime}>\text{now}} (\text{Screening}) \\ \bowtie_{\text{Screening.coordinator_id}=\text{Staff.id}} \text{Staff} \\ \bowtie_{\text{Screening.film_id}=\text{Film.id}} \text{Film} \Big) \Big) \end{aligned}$$

6. Number of screenings for each film

$$\mathcal{T}_{\text{screening_count DESC, title}} \Big(\text{id, title} \bowtie \text{COUNT Screening.id AS screening_count} \Big(\text{Film} \bowtie_{\text{Film.id}=\text{Screening.film_id}} \text{Screening} \Big) \Big)$$

7. Average score for each film

$$\text{id, title} \bowtie \text{ROUND(AVERAGE score, 2) AS average_score} (\text{Film} \bowtie_{\text{Film.id}=\text{Evaluation.film_id}} \text{Evaluation})$$

8. Films ranked by average score

$$\mathcal{T}_{\text{average_score DESC, title}} \Big(\text{id, title} \bowtie \text{ROUND(AVERAGE score, 2) AS average_score} \Big(\text{Film} \bowtie_{\text{Film.id}=\text{Evaluation.film_id}} \text{Evaluation} \Big) \Big)$$

9. Films without evaluations

$$\pi_{\text{Film.id, Film.title}} \Big(\sigma_{\text{Evaluation.film_id IS NULL}} \Big(\text{Film} \bowtie_{\text{Film.id}=\text{Evaluation.film_id}} \text{Evaluation} \Big) \Big)$$

10. Judges with more than five evaluations

$$\begin{aligned} \mathcal{T}_{\text{films_evaluated}} \text{ DESC } & \left(\sigma_{\text{films_evaluated} > 5} \left(\text{id, name, specialization} \right. \right. \\ & \quad \mathfrak{S} \text{ COUNT film_id AS films_evaluated } \left(\text{Judge} \right. \\ & \quad \quad \mathfrak{N}_{\text{Judge.id=Evaluation.judge_id}} \text{ Evaluation} \left. \right) \left. \right) \end{aligned}$$

11. Films with multiple directors

$$\begin{aligned} \sigma_{\text{director_count} > 1} & \left(\text{id, title } \mathfrak{S} \text{ COUNT DISTINCT director_id AS director_count } \left(\right. \right. \\ & \quad \text{Film } \mathfrak{N}_{\text{Film.id=FilmDirector.film_id}} \text{ FilmDirector} \left. \right) \left. \right) \end{aligned}$$

12. Actors appearing in more than two films

$$\begin{aligned} \sigma_{\text{film_count} > 2} & \left(\text{id, name } \mathfrak{S} \text{ COUNT actor_id AS film_count } \left(\right. \right. \\ & \quad \text{Actor } \mathfrak{N}_{\text{Actor.id=FilmActor.actor_id}} \text{ FilmActor} \left. \right) \left. \right) \end{aligned}$$

13. Screenings on a specific date

$$\begin{aligned} \pi_{\text{title, time, room, capacity, name}} & \left(\sigma_{\text{date}=x} (\text{Screening}) \right. \\ & \quad \mathfrak{N}_{\text{Screening.coordinator_id=Staff.id}} \text{ Staff} \\ & \quad \left. \mathfrak{N}_{\text{Screening.film_id=Film.id}} \text{ Film} \right) \end{aligned}$$

14. Number of screenings in each room on a specific date

$$\text{room } \mathfrak{S} \text{ COUNT * AS screening_count } (\sigma_{\text{date}=x} (\text{Screening}))$$

15. Judges without evaluations

$$\begin{aligned} \mathcal{T}_{\text{name}} & \left(\pi_{\text{Judge.id, Judge.name, Judge.specialization}} \left(\sigma_{\text{Evaluation.judge_id IS NULL}} \left(\text{Judge} \right. \right. \right. \\ & \quad \left. \left. \mathfrak{N}_{\text{Judge.id=Evaluation.judge_id}} \text{ Evaluation} \right) \right) \left. \right) \end{aligned}$$

16. Films without screenings

$$\begin{aligned} \pi_{\text{Film.id, Film.title}} & \left(\sigma_{\text{Screening.film_id IS NULL}} \left(\right. \right. \\ & \quad \text{Film } \mathfrak{N}_{\text{Film.id=Screening.film_id}} \text{ Screening} \left. \right) \left. \right) \end{aligned}$$